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Collapsible rotor assembly of a vertical axis wind turbine

Abstract

A collapsible rotor assembly and a method to assemble the rotor assembly are disclosed. The rotor assembly may include a mast, a blade, a first arm and a second arm. The rotor assembly may further include a first connection member attached to a mast top portion and a second connection member attached to a mast bottom portion. The rotor assembly may additionally include a third connection member attached to a blade top portion and a fourth connection member attached to a blade bottom portion. A first arm distal end may be pivotally attached with the first connection member and a first arm proximal end may be pivotally attached with the third connection member. Furthermore, a second arm distal end may be removably attached with the second connection member and a second arm proximal end may be pivotally attached with the fourth connection member.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a rotor assembly of a vertical axis wind turbine, and more specifically to a collapsible rotor assembly of a vertical axis wind turbine and a method for assembling the collapsible rotor assembly.

BACKGROUND

(2) A vertical axis wind turbine includes a plurality of rotor assembly units including, but not limited to, a mast, blades, arms, multiple connection members, and/or the like. A conventional method to install a vertical axis wind turbine includes manufacturing the rotor assembly units in a manufacturing plant/shop, and then transporting the units to the field. Contractors in the field then assemble each unit, one-by-one, to install the wind turbine in the field.

(3) Assembling different rotor assembly units in the field may be challenging and time-consuming for the contractors. For example, it may take more than a day to assemble all the rotor assembly units in the field, which may result in inconvenience for the contractors.

- (4) Thus, there is a need for a rotor assembly that may be easy to install in the field.
- (5) It is with respect to these and other considerations that the disclosure made herein is presented.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The detailed description is set forth with reference to the accompanying drawings. The use of the same reference numerals may indicate similar or identical items. Various embodiments may utilize elements and/or components other than those illustrated in the drawings, and some elements and/or components may not be present in various embodiments. Elements and/or components in the figures are not necessarily drawn to scale. Throughout this disclosure, depending on the context, singular and plural terminology may be used interchangeably.
- (2) FIG. 1 depicts an example vertical axis wind turbine in accordance with the present disclosure.
- (3) FIG. 2 depicts a side view of an example collapsible rotor assembly of a vertical axis wind turbine in accordance with the present disclosure.
- (4) FIG. 3 depicts a flow diagram of a method for bundling a collapsible rotor assembly in accordance with the present disclosure.
- (5) FIGS. 4A and 4B depict a flow diagram of a method for installing the collapsible rotor assembly of FIG. 2 in a field in accordance with the present disclosure.
- (6) FIG. 5 depicts an example snapshot of the collapsible rotor assembly of FIG. 2 being placed on a drive shaft in accordance with the present disclosure.
- (7) FIG. 6 depicts an example snapshot of the collapsible rotor assembly of FIG. 2 placed on the drive shaft in accordance with the present disclosure.
- (8) FIG. 7 depicts an example snapshot of a blade extended away from a mast in accordance with the present disclosure.
- (9) FIG. 8 depicts an example snapshot of a first vertical bracing member connected to the mast in accordance with the present disclosure.
- (10) FIG. 9 depicts an example snapshot of a lower arm connected to the mast in accordance with the present disclosure.
- (11) FIG. 10 depicts an example snapshot of the collapsible rotor assembly of FIG. 2 in an installed state in accordance with the present disclosure.

DETAILED DESCRIPTION

Overview

- (12) The present disclosure is directed to a rotor assembly of a vertical axis wind turbine. Specifically, the present disclosure is directed to a collapsible rotor assembly that may be pre-assembled in a manufacturing plant/shop and transported to field, where a contractor may conveniently install the rotor assembly over a drive shaft or a tower. The rotor assembly may include a plurality of units including, but not limited to, a mast, blades, upper arms, lower arms, upper vertical bracing members, lower vertical bracing members, a plurality of connection members, and/or the like. In an exemplary aspect, the rotor assembly may include three blades, three upper arms and three lower arms. Each blade may be attached to one upper arm and one lower arm. Proximal end of each upper arm may be pivotally attached to a blade top portion and the distal end of the upper arm may be pivotally attached to a mast top portion. Further, proximal end of each lower arm may be pivotally attached to a blade bottom portion and the distal end of the lower arm may be removably attached to a mast bottom portion. Specifically, when the rotor assembly may be in “collapsed state”, the lower arm distal end may be detached from (or not attached with) the mast bottom portion. On the other hand, when the rotor assembly may be in “installed state” in the field, the lower arm distal end may be attached with the mast bottom portion.

(13) Proximal end of each upper vertical bracing member may be pivotally attached to the upper arm between an upper arm proximal and an upper arm distal end. The distal end of the upper vertical bracing member may be removably attached to the mast between a mast top end and a mast bottom end. Specifically, in the rotor assembly collapsed state, the distal end of the upper vertical bracing member may be detached from (or not attached with) the mast. On the other hand, in the rotor assembly installed state, the distal end of the upper vertical bracing member may be attached to the mast.

(14) Similarly, proximal end of each lower vertical bracing member may be pivotally attached to the lower arm between a lower arm proximal and a lower arm distal end. The distal end of the lower vertical bracing member may be removably attached to the mast between the mast top end and the mast bottom end. Specifically, in the rotor assembly collapsed state, the distal end of the lower vertical bracing member may be detached from (or not attached with) the mast. On the other hand, in the rotor assembly installed state, the distal end of the lower vertical bracing member may be attached to the mast.

(15) The present disclosure further describes a method to bundle a collapsible rotor assembly, e.g., in the manufacturing plant/shop before the rotor assembly may be transported to the field. In some aspects, to bundle the rotor assembly, a contractor or a robot may hook the lower arm and the lower vertical bracing member together via cords, ropes, etc. or fasteners to form a first arm-brace bundle. The contractor may then hook the first arm-brace bundle with the blade to form a first blade-arm bundle.

(16) Furthermore, the contractor may hook the upper arm with the upper vertical bracing member to form a second arm-brace bundle. The contractor may then hook the second arm-brace bundle with the blade to form a second blade-arm bundle. In some aspects, the contractor may hook the second arm-brace bundle with the blade such that the first blade-arm bundle may be disposed between the second arm-brace bundle and the blade.

(17) When the second blade-arm bundle may be formed, the rotor assembly may be in the collapsed state, and the contractor may transport the rotor assembly to the field.

(18) The present disclosure additionally describes a method to install the collapsible rotor assembly in the field. To install the rotor assembly, the contractor may attach the mast bottom end to the drive shaft/tower top end. The contractor may then unhook the second blade-arm bundle and extend the blade away from the mast. Further, the contractor may unhook the second arm-brace bundle and attach the upper vertical bracing member to the mast. The contractor may then unhook the first blade arm bundle and attach the lower arm to the mast. Furthermore, the contractor may unhook the first arm brace bundle and attach the lower vertical bracing member to the mast. In this manner, the contractor may install the rotor assembly in the field.

(19) The present disclosure discloses a collapsible rotor assembly of a vertical axis wind turbine that is easy to transport from the shop and convenient for the contractor to install in the field. Since the rotor assembly is “pre-assembled” in the shop, considerable time and effort for the contractor may be saved to install the rotor assembly in the field. Further, the contractor may easily install the rotor assembly by using limited equipment as complex rotor component assembly may not be required to be done in the field, thus saving equipment and logistics cost for the contractor.

(20) These and other advantages of the present disclosure are provided in detail herein.

Illustrative Embodiments

(21) The disclosure will be described more fully hereinafter with reference to the accompanying drawings, in which example embodiments of the disclosure are shown, and not intended to be limiting.

(22) FIG. 1 depicts an example vertical axis wind turbine **100** (wind turbine **100**) in accordance with the present disclosure. The wind turbine **100** may include a rotor assembly having one or more units including, but not limited to, a vertical mast **102**, one or more blades **104**, upper arms **106**, lower arms **108**, upper vertical bracing members **110** and lower vertical bracing members **112**. The

rotor assembly may be a “collapsible” rotor assembly that may be packaged or assembled in a collapsed state (as shown in FIG. 2) in a manufacturing plant/shop and transported to field. In the field, the rotor assembly in the collapsed state may be placed over a drive shaft **114** and/or a tower **116** and then easily “extended” to be in a final installed state, as shown in FIG. 1. Structural details of the collapsible rotor assembly are depicted in FIG. 2 and described later in the description below. Further, the process of installing the collapsible rotor assembly in the field is described later below in conjunction with FIGS. 4A and 4B.

(23) The mast **102** may be made of metal such as aluminum, iron, steel, lightweight alloy, and/or the like. In the installed state, the mast bottom end may be connected to the drive shaft **114** (specifically to a drive shaft top end). Further, in the installed state, the mast **102** may be disposed or aligned perpendicular to ground. The mast **102** may be of any length depending on one or more parameters including wind turbine dimensions, location of geographical area where the wind turbine **100** may be located, desired electric power output from the wind turbine **100**, and/or the like. Further, the mast **102** may be polygonal in shape having a predefined count of faceted surfaces. For example, in a preferred aspect, the mast **102** may have hexagonal cross-section. In other aspects, the mast **102** may have cuboidal, pentagonal, octagonal, etc., cross-sectional shape. Furthermore, in some aspects, the mast **102** may be hollow throughout a mast length. In other aspects, one or more mast interior portions may be hollow, and remaining mast interior portions may be solid.

(24) The drive shaft bottom end may be connected directly or through a gearbox to a power generator (not shown) that may be configured to generate electric power when the mast **102** rotates (thus rotating the drive shaft **114**). Specifically, the power generator may convert mast kinetic energy into electric energy, thus generating electric power from the wind turbine **100**.

(25) In an exemplary aspect, the wind turbine **100** may include three blades **104** that may be disposed parallel to a mast longitudinal axis. The blades **104** may be hollow elongated airfoil shaped blades, having lengths that may depend on the wind turbine dimensions. The blades **104** may be lightweight, and made of metal such as aluminum. In other aspects, the blades **104** may be made of any other material, e.g., fiberglass, plastic, etc.

(26) In some aspects, the wind turbine **100** may include three upper arms **106** and three lower arms **108**. A pair of upper and lower arms may connect a blade to the mast **102**. Stated another way, each blade may be attached to one upper arm and one lower arm, and respective upper and lower arms may be connected to the mast **102**. In some aspects, one or more connections between the upper and lower arms **106**, **108** and the blades **104**, and between the upper and lower arms **106**, **108** and the mast **102** may be pivotal and/or removable. Details of the arm-blade connections (via respective arm-blade connection members) and the arm-mast connections (via respective arm-mast connection members) are described later below in conjunction with FIG. 2.

(27) Each of the upper and lower arms **106**, **108** may be made of the same or similar material as the mast **102** and/or the blades **104**, and may be cuboidal in shape. Specifically, the upper and lower arms **106**, **108** may be hollow elongated cuboids having dimensions (e.g., length and width) that may vary based on the wind turbine dimensions. In other aspects, the upper and lower arms **106**, **108** may be made of extruded or pultruded material and may be airfoil shaped, which may make the upper and lower arms **106**, **108** aerodynamic.

(28) In operation, wind may cause the blades **104** to rotate axially relative to the mast **102** longitudinal axis due to the blade airfoil shape. Blade rotation may cause upper and lower arm movement, and thus cause mast rotation. Mast rotation may enable the power generator to generate electric power, as described above.

(29) To ensure that wind turbine structure is robust and can withstand wind turbulence, the wind turbine **100** may include the upper and lower vertical bracing members **110**, **112**. One end of the upper vertical bracing member **110** may be connected to the upper arm **106** at a point between an upper arm proximal end and an upper arm distal end. The other end of the upper vertical bracing

member **110** may be connected to the mast **102** at a point between a mast top end and a mast bottom end. Similarly, one end of the lower vertical bracing member **112** may be connected to the lower arm **108** at a point between a lower arm proximal end and a lower arm distal end. The other end of the lower vertical bracing member **112** may be connected to the mast **102** at a point between the mast top end and the mast bottom end. Connections of the upper and lower vertical bracing members **110**, **112** with the upper and lower arms **106**, **108** (via respective bracing member-arm connection members) and connections of the upper and lower vertical bracing members **110**, **112** with the mast **102** (via respective bracing member-mast connection members) may be pivotal and/or removable. Structural details of connections between the upper and lower vertical bracing members **110**, **112** with the upper and lower arms **106**, **108**, and between the upper and lower vertical bracing members **110**, **112** with the mast **102** are described below in conjunction with FIG. 2.

(30) In some aspects, the upper and lower vertical bracing members **110**, **112** may be made of same material as the blade **104** and/or the mast **102**. Further, the upper and lower vertical bracing members **110**, **112** may be shaped as hollow cuboids, and may have dimensions (e.g., length and width) that may be based on the wind turbine dimensions.

(31) The upper and lower vertical bracing members **110**, **112**, as depicted in FIG. 1 and described above, may enhance wind turbine structural integrity. Specifically, a person ordinarily skilled in the art may appreciate that the upper and lower arms **106**, **108** may experience gravitational force, which may exert stresses on the connections between the upper and lower arms **106**, **108** and the blades **104**, and between the upper and lower arms **106**, **108** and the mast **102**. Attachment of the upper and lower vertical bracing members **110**, **112**, as depicted in FIG. 1, may provide vertical support to the upper and lower arms **106**, **108** and may facilitate in reducing stresses at the connections described above due to gravitational force and allow for blade arms with smaller profiles to the wind (and thus less drag).

(32) FIG. 2 depicts a side view of an example collapsible rotor assembly **200** of the vertical axis wind turbine **100** in accordance with the present disclosure. The rotor assembly **200** may be same as the rotor assembly described above in conjunction with FIG. 1. In some aspects, FIG. 2 depicts the rotor assembly **200** in a “collapsed” state. Although FIG. 2 depicts a single blade, the rotor assembly **200** may include three blades as described above.

(33) The rotor assembly **200** may include a mast **202** having a mast top portion **204** and a mast bottom portion **206**. The mast **202** may be same as the mast **102**. The rotor assembly **200** may further include a blade **208** (or three blades **208**) having a blade top portion **210** and a blade bottom portion **212**. The blade **208** may be same as the blade **104**.

(34) The rotor assembly **200** may further include a first connection member **214** attached to the mast top portion **204** and a second connection member **216** attached to the mast bottom portion **206**. The rotor assembly **200** may additionally include a third connection member **218** attached to the blade top portion **210** and a fourth connection member **220** attached to the blade bottom portion **212**.

(35) Further, the rotor assembly **200** may include a first arm **222** (or an upper arm **222**) and a second arm **224** (or a lower arm **224**). The upper and lower arms **222**, **224** may be same as the upper and lower arms **106**, **108** described above in conjunction with FIG. 1. The upper arm **222** may include an upper arm proximal end **226** and an upper arm distal end **228**, and the lower arm **224** may include a lower arm proximal end **230** and a lower arm distal end **232**.

(36) The upper arm proximal end **226** may be pivotally connected (e.g., via a pinned/hinged connection) with the third connection member **218** and the upper arm distal end **228** may be pivotally connected (e.g., via a pinned/hinged connection) with the first connection member **214**. Stated another way, the third connection member **218** may enable pivotal connection between the upper arm **222** and the blade top portion **210**, and the first connection member **214** may enable pivotal connection between the upper arm **222** and the mast top portion **204**. In some aspects, the

first connection member **214** and the third connection member **218** may enable fixed or “non-removable” mast-arm and blade-arm connections, respectively. Stated another way, the upper arm **222** may be pivotally and “fixedly” connected between the blade top portion **210** and the mast top portion **204**.

(37) The lower arm proximal end **230** may be pivotally connected (e.g., via a pinned/hinged connection) with the fourth connection member **220**. Stated another way, the fourth connection member **220** may enable pivotal connection between the lower arm **224** and the blade bottom portion **212**. In some aspects, the fourth connection member **220** may enable fixed or “non-removable” blade-arm connection. Stated another way, the lower arm **224** may be pivotally and “fixedly” connected to the blade bottom portion **212**.

(38) The lower arm distal end **232** may be configured to be removably attached with the second connection member **216**. In some aspects, in the collapsed state of the rotor assembly **200**, the lower arm distal end **232** may be detached from (or not connected with) the second connection member **216**, as shown in FIG. 2. On the other hand, when the rotor assembly **200** may be installed in the field, the lower arm distal end **232** may be attached with the second connection member **216**. The second connection member **216** may enable pivotal connection (e.g., via a pin/hinge connector) or non-pivotal connection (e.g., via bolts) between the lower arm distal end **232** and the mast bottom portion **206**.

(39) The rotor assembly **200** may further include a brace connection member **234** attached to the mast **202** between the mast top portion **204** and the mast bottom portion **206**. In the exemplary aspect depicted in FIG. 2, the brace connection member **234** may be equidistant from the mast top portion **204** and the mast bottom portion **206**. In other aspects, the brace connection member **234** may be closer to one of the mast portions, i.e., the mast top portion **204** or the mast bottom portion **206**.

(40) The rotor assembly **200** may additionally include a fifth connection member **236** attached to the upper arm **222** and a sixth connection member **238** attached to the lower arm **224**. The fifth connection member **236** may be attached to the upper arm **222** anywhere between the upper arm proximal end **226** and the upper arm distal end **228**. Similarly, the sixth connection member **238** attached to the lower arm **224** anywhere between the lower arm proximal end **230** and the lower arm distal end **232**.

(41) Furthermore, the rotor assembly **200** may include a first bracing member **240** (or an upper vertical bracing member **240**) and a second bracing member **242** (or a lower vertical bracing member **242**) that may be same as the upper and lower vertical bracing members **110**, **112** described above in conjunction with FIG. 1. The upper vertical bracing member **240** may include an upper member proximal end **244** and an upper member distal end **246**, and the lower vertical bracing member **242** may include a lower member proximal end **248** and a lower member distal end **250**.

(42) The upper member proximal end **244** may be pivotally connected with the upper arm **222** between the upper arm proximal end **226** and the upper arm distal end **228** via the fifth connection member **236**. Stated another way, the fifth connection member **236** may enable a pivotal connection (e.g., via a pin/hinge connector) between the upper arm **222** and the upper member proximal end **244**. In some aspects, the connection between the upper arm **222** and the upper member proximal end **244** may be fixed or non-removable.

(43) The upper member distal end **246** may be configured to be removably attached with the brace connection member **234** (e.g., via bolts or a pin/hinge connector). Stated another way, the upper member distal end **246** may be configured to be removably attached to the mast **102** via the brace connection member **234**. In an exemplary aspect, when the rotor assembly **200** may be in the collapsed state, the upper member distal end **246** may be detached from (or not connected with) the brace connection member **234**, as shown in FIG. 2. However, in the installed state of the rotor assembly **200**, the upper member distal end **246** may be attached to the brace connection member

234/mast 102, as shown in FIGS. **8-10** and described later in detail below.

(44) In a similar manner, the lower member proximal end **248** may be pivotally connected with the lower arm **224** between the lower arm proximal end **230** and the lower arm distal end **232** via the sixth connection member **238**. Stated another way, the sixth connection member **238** may enable a pivotal connection (e.g., via a pin/hinge connector) between the lower arm **224** and the lower member proximal end **248**. In some aspects, the connection between the lower arm **224** and the lower member proximal end **248** may be fixed or non-removable.

(45) The lower member distal end **250** may be configured to be removably attached with the brace connection member **234** (e.g., via bolts or a pin/hinge connector). Stated another way, the lower member distal end **250** may be configured to be removably attached to the mast **102** via the brace connection member **234**. In an exemplary aspect, when the rotor assembly **200** may be in the collapsed state, the lower member distal end **250** may be detached from (or not connected with) the brace connection member **234**, as shown in FIG. **2**. However, in the installed state of the rotor assembly **200**, the lower member distal end **250** may be attached to the brace connection member **234/mast 102**, as shown in FIG. **10** and described later in detail below.

(46) In some aspects, a wind turbine manufacturer or contractor may “hook” or “bundle” different rotor assembly components/units described above to assemble the rotor assembly **200** in the collapsed state, as shown in FIG. **2**. The contractor may then transport the rotor assembly **200** in the collapsed state to the field, where the contractor may place the collapsed rotor assembly **200** on the drive shaft **114/tower 116**, and then start to “unhook” each component one-by-one to conveniently install the rotor assembly **200** in the field. A person ordinarily skilled in the art may appreciate that since the rotor assembly **200** in the collapsed state is “pre-assembled”, the contractor may not be required to assemble the rotor assembly **200** in the field, thereby saving considerable time and effort for the contractor.

(47) The process of bundling the rotor assembly components to assemble the rotor assembly **200** in the collapsed state is described below in conjunction with FIG. **3**, and the process of installing the rotor assembly **200** in the field is described later below in conjunction with FIGS. **4A, 4B** and **5-10**.

(48) FIG. **3** depicts a flow diagram of a method **300** for bundling the collapsible rotor assembly **200** in accordance with the present disclosure. FIG. **3** may be described with continued reference to prior figures, including FIGS. **1** and **2**. The following process is exemplary and not confined to the steps described hereafter. Moreover, alternative embodiments may include more or less steps that are shown or described herein and may include these steps in a different order than the order described in the following example embodiments.

(49) In some aspects, before bundling the collapsible rotor assembly **200** as described below, the contractor may attach the plurality of rotor assembly components that are described above in the manufacturing plant/shop. For example, the contractor (or a robot) may attach the first and second connection members **214, 216** to the mast top and bottom portions **204, 206** respectively. The contractor may further attach the third and fourth connection members **218, 220** to the blade top and bottom portions **210, 212** respectively. The contractor may additionally attach the fifth and sixth connection members **236, 238** to the upper and lower arms **222, 224** respectively, as described above. The contractor may also attach the brace connection member **234** to the mast **102** (e.g., at a mast center portion).

(50) The contractor may then pivotally attach the upper member proximal end **244** to the fifth connection member **236** and pivotally attach the lower member proximal end **248** to the sixth connection member **238**. The contractor may further pivotally attach the lower arm proximal end **230** to the fourth connection member **220**, and attach the upper arm proximal end **226** to the third connection member **218**.

(51) In some aspects, the contractor may further pivotally attach the upper arm distal end **228** to the first connection member **214** in the manufacturing plant/shop. In other aspects, the contractor may not attach the upper arm distal end **228** to the first connection member **214** in the manufacturing

plant/shop, but may instead attach the upper arm distal end **228** to the first connection member **214** in the field.

(52) The contractor may make the attachments described above for all the three blade assemblies. When the connections/attachments, as described above, are made, the contractor (or the robot) may perform the steps depicted in FIG. 3 and described below to “bundle” the rotor assembly **200** and make it ready for transportation to the field.

(53) The method **300** starts at step **302**. At step **304**, the method **300** may include hooking the lower arm **224** with the lower vertical bracing member **242** to form a first arm-brace bundle (or a lower arm-lower vertical bracing member bundle). In some aspects, the contractor (or the robot) may hook the lower arm **224** with the lower vertical bracing member **242** by tying the lower arm **224** and the lower vertical bracing member **242** together with cords, ropes, etc., or via removable mechanical fasteners (not shown). In some aspects, the lower arm **224** and the lower vertical bracing member **242** may be disposed parallel to each other in the first arm-brace bundle, as shown in FIG. 2.

(54) At step **306**, the method **300** may include hooking the first arm-brace bundle with the blade **208** to form a first blade-arm bundle (or blade-lower arm bundle). In some aspects, the contractor (or the robot) may hook the first arm-brace bundle with the blade **208** by using cords, ropes, etc., or removable mechanical fasteners (not shown).

(55) At step **308**, the method **300** may include hooking the upper arm **222** with the upper vertical bracing member **240** to form a second arm-brace bundle (or an upper arm-upper vertical bracing member bundle). In some aspects, the contractor (or the robot) may hook the upper arm **222** with the upper vertical bracing member **240** by using cords, ropes, etc., or removable mechanical fasteners (not shown). Further, in the second arm-brace bundle, the upper arm **222** and the upper vertical bracing member **240** may be disposed parallel to each other.

(56) At step **310**, the method **300** may include hooking the second arm-brace bundle with the blade **208** to form a second blade-arm bundle. As shown in FIG. 2, the second arm-brace bundle may be hooked with the blade **208** such that the first arm-blade bundle may be disposed between the second arm-brace bundle and the blade **208**.

(57) When the second blade-arm bundle may be formed, the rotor assembly **200** may take the shape/arrangement as shown in FIG. 2. Stated another way, the rotor assembly **200** may be disposed in the collapsed state when the steps described above may be performed.

(58) Responsive to performing the steps described above, the contractor (or the robot) may laterally or horizontally place the rotor assembly **200** in a truck, such that the mast longitudinal axis may be parallel to truck cargo bed. The contractor may then transport the rotor assembly **200** to the field. When the rotor assembly **200** reaches the field, the rotor assembly **200** may be installed on the drive shaft **114**/tower **116**, as described below in conjunction with FIGS. 4A, 4B, 5-10.

(59) At step **312**, the method **300** ends.

(60) FIGS. 4A and 4B depict a flow diagram of a method **400** for installing the collapsible rotor assembly **200** in the field in accordance with the present disclosure. FIGS. 4A and 4B may be described with continued reference to prior figures, including FIGS. 1-3. Further, FIGS. 4A and 4B may be described with continued reference to subsequent figures, including FIGS. 5-10. The following process is exemplary and not confined to the steps described hereafter. Moreover, alternative embodiments may include more or less steps that are shown or described herein and may include these steps in a different order than the order described in the following example embodiments.

(61) The method **400** starts at step **402**. At step **404**, the method **400** may include attaching the mast bottom end to the drive shaft **114** or the tower **116**, as shown in FIG. 5. Specifically, at step **404**, the contractor may attach a lifting means (e.g., a crane or crane hook) to a mast top end or a point in proximity to the mast top portion **204** and lift the rotor assembly **200** from the truck. Responsive to lifting the rotor assembly **200** from the truck, the contractor may dispose, via the lifting means, the

rotor assembly **200** on top of the drive shaft **114** or the tower **116**, such that a tower longitudinal axis may be parallel to the mast longitudinal axis. The contractor may then dispose the mast bottom end in proximity to the drive shaft **114** or the tower top end, and attach the mast bottom end to the drive shaft **114** or the tower **116** via fastening means (e.g., bolts, not shown). When the mast bottom end may be attached to the drive shaft **114**, the rotor assembly **200** may be attached to the drive shaft **114**/tower **116**, as shown in FIG. 6.

(62) At step **406**, the method **400** may include unhooking the second blade-arm bundle. As described above in conjunction with FIG. 3, the second blade-arm bundle may include the second arm-brace bundle and the blade **208**. Further, as described above, the second arm-brace bundle may include the upper arm **222** and the upper vertical bracing member **240**. In some aspects, the contractor (or the robot) may unhook the second blade-arm bundle by unhooking respective cords, ropes or the removable mechanical fasteners, as described above in conjunction with FIG. 3.

(63) At step **408**, the method **400** may include laterally extending the blade **208** away from the mast **202** such that the blade **208** may be disposed at a predefined distance from the mast **202**. The predefined distance may depend on the wind turbine dimensions. In some aspects, at the step **408**, the contractor may remove the lifting means from the mast top end and attach the lifting means to the blade top end. The contractor may then extend, via the lifting means, the blade **208** away from the mast **202**. An example view of the blade **208** extended away from the mast **202** is depicted in FIG. 7.

(64) At step **410**, the method **400** may include unhooking the second arm-brace bundle. As described above in conjunction with FIG. 3, the second arm-brace bundle may include the upper arm **222** with the upper vertical bracing member **240**. The contractor may unhook the second arm-brace bundle by unhooking respective cords, ropes, etc. or the mechanical fasteners.

(65) At step **412**, the method **400** may include axially rotating the upper vertical bracing member **240** about the upper member proximal end **244** (or about the fifth connection member **236**), as shown in FIG. 7. At step **414**, the method **400** may include attaching (e.g., via bolts, pinned/hinged connectors, etc.) the upper vertical bracing member **240**, or specifically the upper member distal end **246**, to the mast **202** between the mast top portion **204** (or mast top end) and the mast bottom portion **206** (or mast bottom end). Specifically, at the step **414**, the upper member distal end **246** may be attached to the brace connection member **234**.

(66) An example view of the upper vertical bracing member **240** connected with the mast **202** is depicted in FIG. 8. Responsive to attaching the upper vertical bracing member **240** with the mast **202**, the contractor may remove the lifting means from the blade top end and attach the lifting means to the lower arm distal end **232**. Responsive to attaching the lifting means to the lower arm distal end **232**, the method **400** moves to step **416**. At the step **416**, the method **400** may include unhooking the first blade-arm bundle. As described above in conjunction with FIG. 3, the first blade-arm bundle may include the first arm-brace bundle and the blade **208**. Further, the first arm-brace bundle may include the lower arm **224** and the lower vertical bracing member **242**. In some aspects, the contractor (or the robot) may unhook the first blade-arm bundle by unhooking respective cords, ropes or the removable mechanical fasteners, as described above in conjunction with FIG. 3.

(67) At step **418**, the method **400** may include axially rotating the lower arm **224** about the lower arm proximal end **230** by using the lifting means. Specifically, at the step **418**, the contractor may use the lifting means to axially rotate the lower arm **224** about the fourth connection member **220**, as shown in FIG. 8.

(68) At step **420**, the method **400** may include attaching (e.g., via bolts, pin/hinge connector, etc.) the lower arm distal end **232** to the mast **202** via the second connection member **216**. Specifically, at step **420**, the lower arm distal end **232** may be attached to the second connection member **216**. An example view of the lower arm distal end **232** or the lower arm **224** connected to the second connection member **216** is depicted in FIG. 9.

(69) When the lower arm distal end **232** may be attached to the second connection member **216**, the lifting means may be removed from the lower arm distal end **232** and attached to the lower member distal end **250**.

(70) At step **422**, the method **400** may include unhooking the first arm-brace bundle. At step **424**, the method **400** may include lifting the lower member distal end **250** by using the lifting means and axially rotating the lower vertical bracing member **242** about the lower member proximal end **248**. Specifically, at step **424**, the contractor may axially rotate the lower vertical bracing member **242** about the sixth connection member **238**, as shown in FIG. **9**.

(71) At step **426**, the method **400** may include attaching the lower member distal end **250** to the mast **202** between the mast top and bottom portions **204**, **206**. Specifically, at step **426**, the contractor may attach the lower member distal end **250** to the brace connection member **234**, and remove the lifting means from the lower member distal end **250**. When the lower member distal end **250** may be attached to the brace connection member **234**, the rotor assembly **200** may be in the “installed state”, as shown in FIG. **10**.

(72) The contractor may repeat the same process as described above for the remaining two blades to install the vertical axis wind turbine **100** in the field.

(73) At step **428**, the method **400** ends.

(74) In the above disclosure, reference has been made to the accompanying drawings, which form a part hereof, which illustrate specific implementations in which the present disclosure may be practiced. It is understood that other implementations may be utilized, and structural changes may be made without departing from the scope of the present disclosure. References in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a feature, structure, or characteristic is described in connection with an embodiment, one skilled in the art will recognize such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

(75) It should also be understood that the word “example” as used herein is intended to be non-exclusionary and non-limiting in nature. More particularly, the word “example” as used herein indicates one among several examples, and it should be understood that no undue emphasis or preference is being directed to the particular example being described.

(76) With regard to the processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted. In other words, the descriptions of processes herein are provided for the purpose of illustrating various embodiments and should in no way be construed so as to limit the claims.

(77) Accordingly, it is to be understood that the above description is intended to be illustrative and not restrictive. Many embodiments and applications other than the examples provided would be apparent upon reading the above description. The scope should be determined, not with reference to the above description, but should instead be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. It is anticipated and intended that future developments will occur in the technologies discussed herein, and that the disclosed systems and methods will be incorporated into such future embodiments. In sum, it should be understood that the application is capable of modification and variation.

(78) All terms used in the claims are intended to be given their ordinary meanings as understood by those knowledgeable in the technologies described herein unless an explicit indication to the

contrary is made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc., should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary. Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments could include, while other embodiments may not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments.

Claims

1. A rotor assembly comprising: a mast having a mast top portion and a mast bottom portion; a blade having a blade top portion and a blade bottom portion; a first connection member attached to the mast top portion and a second connection member attached to the mast bottom portion; a third connection member attached to the blade top portion and a fourth connection member attached to the blade bottom portion; a first arm having a first arm proximal end and a first arm distal end, wherein: the first arm distal end is pivotally attached with the first connection member, and the first arm proximal end is pivotally attached with the third connection member; and a second arm having a second arm proximal end and a second arm distal end, wherein: the second arm distal end is removably attached with the second connection member that is attached to the mast bottom portion, and the second arm proximal end is pivotally attached with the fourth connection member, wherein the rotor assembly is in a collapsed state when the second arm distal end is detached from the second connection member, and the rotor assembly is in an installed state when the second arm distal end is attached to the second connection member, and wherein a blade longitudinal axis is parallel to a mast longitudinal axis in both the collapsed state and the installed state.
2. The rotor assembly of claim 1 further comprising: a first bracing member having a first bracing member proximal end and a first bracing member distal end, wherein: the first bracing member proximal end is pivotally connected with the first arm between the first arm proximal end and the first arm distal end, and the first bracing member distal end is connected with the mast; a second bracing member having a second bracing member proximal end and a second bracing member distal end, wherein: the second bracing member proximal end is pivotally connected with the second arm between the second arm proximal end and the second arm distal end, and the second bracing member distal end is connected with the mast; and a brace connection member attached to the mast between the mast top portion and the mast bottom portion, wherein the first bracing member and the second bracing member are attached to the mast via the brace connection member, wherein the brace connection member is disposed at a mast center portion, equidistant and equally spaced from the mast top portion and the mast bottom portion, wherein the first bracing member distal end is removably connected with the brace connection member, and wherein the first bracing member distal end is detached from the brace connection member in the collapsed state.
3. The rotor assembly of claim 2 further comprising a fifth connection member attached to the first arm between the first arm proximal end and the first arm distal end, wherein the first bracing member proximal end is pivotally connected with the first arm via the fifth connection member.
4. The rotor assembly of claim 2, wherein the second bracing member distal end is removably connected with the brace connection member, and wherein the second bracing member distal end is detached from the brace connection member in the collapsed state.
5. The rotor assembly of claim 4 further comprising a sixth connection member attached to the second arm between the second arm proximal end and the second arm distal end, wherein the second bracing member proximal end is pivotally connected with the second arm via the sixth connection member.

6. The rotor assembly of claim 1, wherein the mast is a polygonal mast.
 7. The rotor assembly of claim 1, the rotor assembly is part of a vertical axis wind turbine.
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